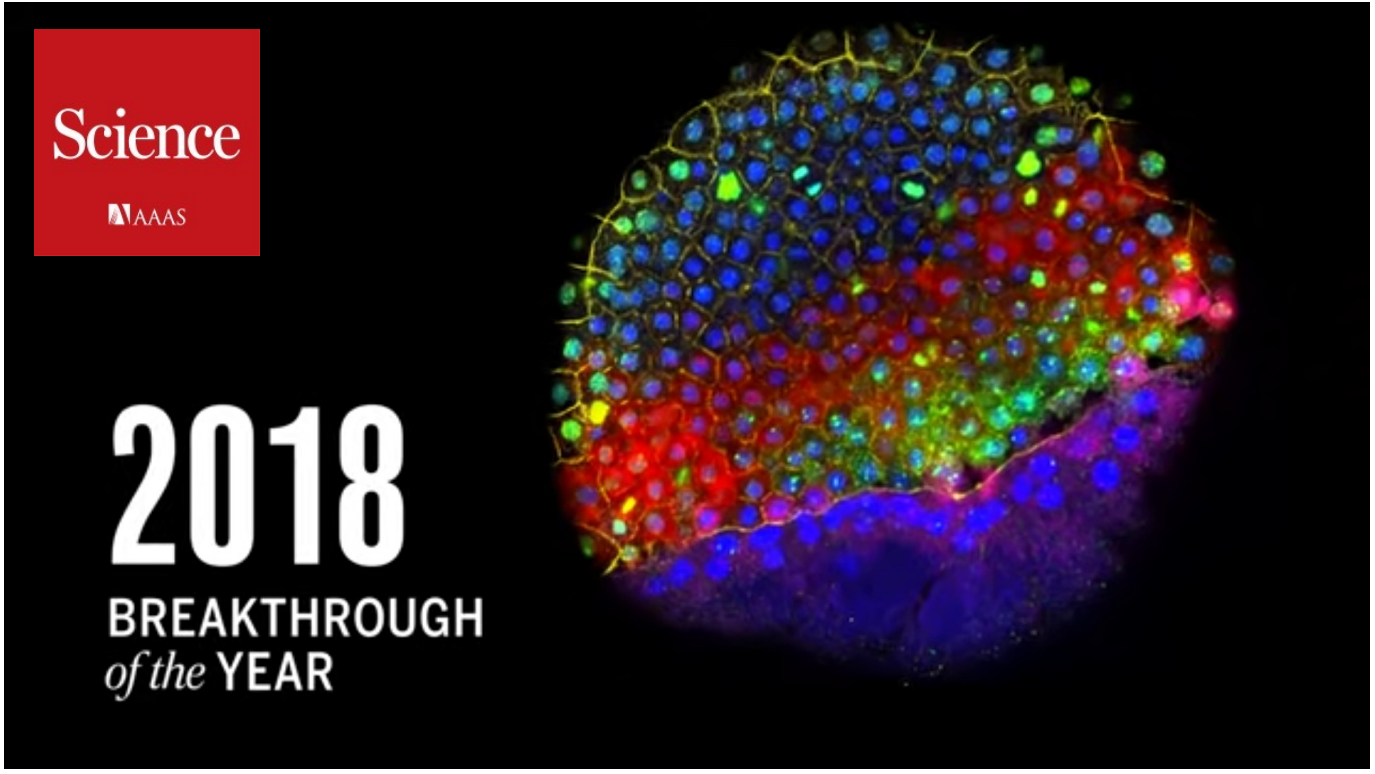
Single Cell RNAseq Analysis

Nikolay Oskolkov, Metabolic Research Group leader at NIRI, Riga, Latvia

PR Stats course, **Session 4: cell-type deconvolution of bulk RNAseq data**



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Nature method of the year:
2019 (single cell multi-omics),
2020 (spatial transcriptomics)

Image adapted from Science, 362, 6421, 2018

Session content

From a single-cell reference atlas to estimated cell-type fractions in bulk tissue

1 Concept

- Why bulk RNA-seq mixes signals from multiple cell types
- Reference profiles from single-cell / single-nucleus atlases
- What a deconvolution estimate means

2 Method

- Signature matrix and mixture model
- DeconRNASeq: non-negative quadratic programming
- Assumptions, identifiability and limitations

3 Application

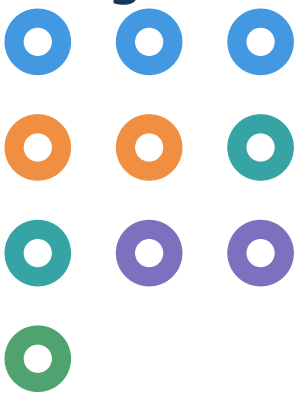
- PredictFiberType: slow vs fast muscle fibres
- Run the published R script on example data
- Compare predictions with measured fractions

Central question: can bulk expression reveal the cellular composition of a tissue?

Why deconvolve bulk RNA-seq?

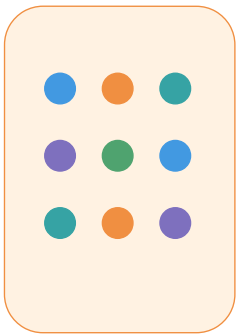
A bulk sample combines transcriptional signals from many cell types

Single-cell atlas



Label cells and derive cell-type expression signatures

Bulk tissue



Expression is a mixture of cell-type contributions

Deconvolution

Estimate the fractions that best explain the observed bulk expression.

Output: one estimated composition per sample, e.g. 35% slow and 65% fast fibres.

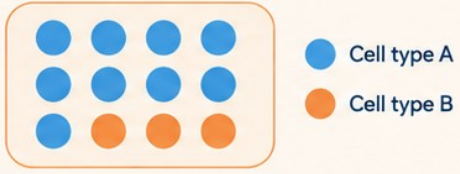
Important: the output is an inferred composition—not a direct count of cells.

Cell composition can bias bulk differential expression

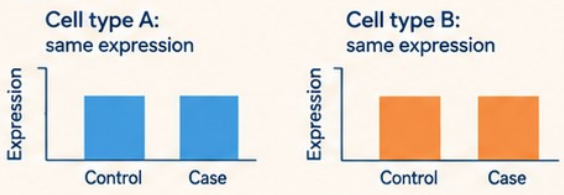
Example: Cell type B is more abundant in cases, so its marker gene appears upregulated.

CONTROL

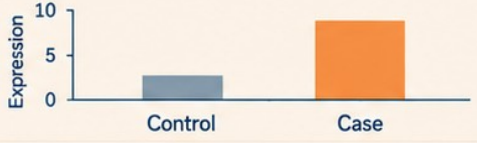
Cell composition: 75% A / 25% B
(9 A cells, 3 B cells)



Within-cell-type gene expression is unchanged

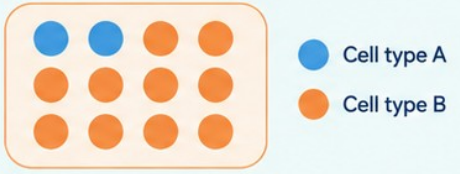


Bulk RNA-seq (marker gene for cell type B)

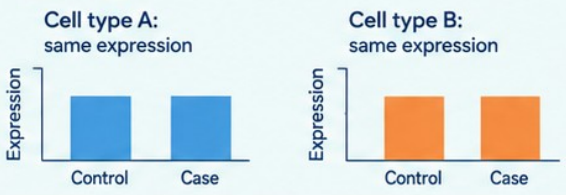


CASE

Cell composition: 25% A / 75% B
(3 A cells, 9 B cells)



Within-cell-type gene expression is unchanged



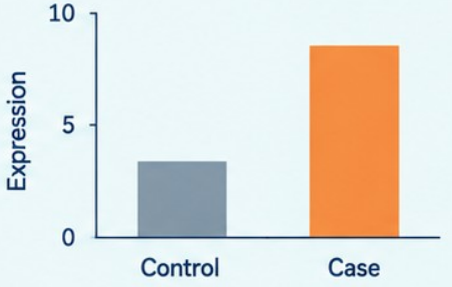
Bulk RNA-seq (marker gene for cell type B)



Apparent upregulation in bulk

Bulk RNA-seq

Differential expression of marker gene (aggregated across all cells)



The increase in bulk expression reflects the higher proportion of cell type B, not a change in expression within cell types.

Bulk DE can reflect a shift in cell proportions—not a change in expression within cells.

The core idea: learn signatures, then estimate mixtures

Reference profiles define what each cell type contributes to bulk expression

1 Reference atlas

Single-cell / single-nucleus RNA-seq

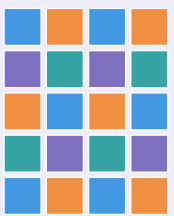
Cell labels → cluster-specific markers → average expression per cell type

2 Signature matrix S

Rows = informative genes

Columns = cell types

Each column describes a reference expression profile



Genes × cell types

3 Bulk mixture X

Observed expression for each bulk sample.

Estimate fractions A such that:

$X \approx S \times A$

The reference and bulk data must use compatible genes, identifiers and expression scales

DeconRNASeq: a constrained mixture model

Gong & Szustakowski, Bioinformatics (2013), 29:1083–1085

Bulk expression is modelled as a weighted sum of reference profiles:

$$X \approx S A$$

- X: observed gene expression in bulk samples
- S: reference signature matrix (genes × cell types)
- A: unknown cell-type proportions to estimate

How are fractions estimated?

- Non-negative least-squares / quadratic programming
- Fractions are constrained to be ≥ 0
- The solution seeks the mixture that best reconstructs the observed bulk profile
- Estimated fractions can be compared with known mixtures or independent assays

Reference: Gong T. & Szustakowski J.D. (2013). doi:10.1093/bioinformatics/btt090

What makes deconvolution work—and fail?

The quality of the reference and specificity of signatures are critical

Good practice

- Use a biologically relevant single-cell atlas
- Ensure cell types are represented in the reference
- Match gene identifiers and expression units
- Select informative, cell-type-specific genes
- Validate against measured proportions when possible

Common pitfalls

- Missing cell types can bias all estimated fractions
- Similar cell types may be hard to separate
- Ref and bulk platforms / conditions may differ
- RNA content per cell differs: RNA fractions need not equal cell fractions
- Fractions are model-based estimates, not ground truth

The model cannot recover information absent from the reference signatures.

Method paper: DeconRNASeq (2013)

A statistical framework for RNA-seq mixture deconvolution

Gong & Szustakowski: "DeconRNASeq: a statistical framework for deconvolution of heterogeneous tissue samples based on mRNA-Seq data"

- Model: bulk expression as weighted sum of tissue / cell-type profiles
- Solver: non-negative quadratic programming
- Validation: in-silico mixtures with known proportions
- Reported correlation with true mixture fractions reached 0.9754 in their benchmark

Schematic workflow





signature matrix

Optimize non-negative mixture weights

PredictFiberType applies this general strategy to slow- and fast-twitch muscle fibres.

Application: PredictFiberType method

Oskolkov et al. (2022), High-throughput muscle fiber typing from RNA sequencing data

Biological question

Can total-RNA bulk sequencing estimate the relative abundance of type I (slow-twitch) and type II (fast-twitch) skeletal muscle fibres?

The method uses cell-type signatures derived from a skeletal-muscle snRNA-seq atlas and applies DeconRNASeq to bulk expression.

Validation reported in the paper

- Compared predictions with ATPase staining and myosin heavy-chain isoform measurements
- Spearman correlation with myosin based composition: r = 0.83
- Correlation with ATPase staining: r = 0.44
- Inference remained feasible at ~10,000 paired-end reads on average

Reference: Skeletal Muscle (2022), doi:10.1186/s13395-022-00299-4

Validating muscle-cell deconvolution against biological ground truth

Two complementary measurements quantify muscle-fibre composition—but measure different properties.

ATPase histochemistry (ATP staining)

Classifies individual fibres by differential myofibrillar ATPase activity after pre-incubation at different pH values

Example: fibre cross-section



Readout: fibre-by-fibre classification and proportions in a tissue section; spatial information is retained.

Myosin heavy-chain (MyHC) isoforms

Measures relative abundance of fibre-type-associated MyHC proteins (commonly I, IIa, IIx; IIb in some species / muscles).

Example: protein bands



Measured by electrophoresis, immuno assay or mass-spectrometry, depending on the protocol. Quantification is often at sample level.

Readout: bulk protein isoform composition; typically less spatially resolved than histochemistry.

How this validates RNA-seq deconvolution

snRNA-seq reference
Cell-type expression signatures

›

Bulk RNA-seq
Mixed transcriptional signal

›

Deconvolution
Estimated fibre-type fractions

›

Independent ground truth
Compare with ATP staining and MyHC

Key distinction: ATP staining classifies individual fibres in tissue; MyHC assays quantify fibre-type-associated proteins. Neither directly measures RNA.

Tutorial: run PredictFiberType in R

Repository: github.com/OlaHanssonLab/PredictFiberType

Input

bulk_expr.txt

Normalized, batch-corrected bulk RNA-seq matrix

Rows = gene symbols; columns = samples

Method

PredictFiberType.R

Uses the published signature matrix and DeconRNASeq method to estimate type I fibertype fractions.

Output

deconv_prediction.txt

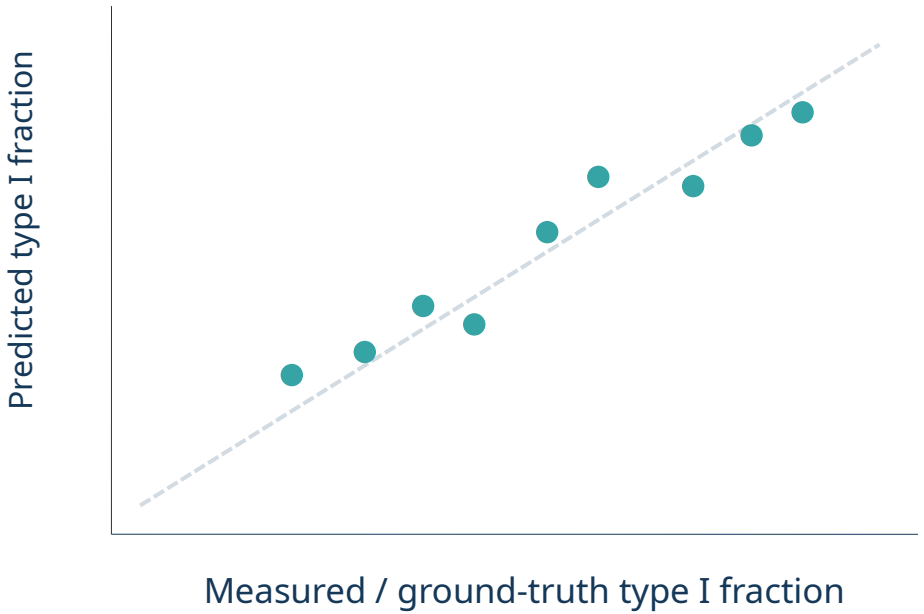
Predicted type I fractions per sample

Optional true_type1.txt enables validation plot / correlation

Run: Rscript PredictFiberType.R bulk_expr.txt true_type1.txt

Assessing predictions against ground truth

A deconvolution result is most informative when independent reference measurements are available



What to report

- Scatter plot: predicted vs measured fractions
- Pearson / Spearman correlation
- Calibration / slope and intercept
- Error metrics (MAE or RMSE)
- Check sample matching, scale and missing values

Correlation measures ranking; it does not by itself guarantee unbiased fractions.

Practical workflow and interpretation

Keep the reference, preprocessing and validation transparent

- 1. Start from normalized, batch-corrected bulk RNA-seq expression; use gene symbols as expected by the script.
- 2. Ensure bulk samples and the truth file have matching sample identifiers.
- 3. Clone the repository and install the Bioconductor dependency DeconRNASeq.
- 4. Run PredictFiberType.R and inspect deconv_prediction.txt.
- 5. If the two classes are exhaustive, type II fraction is approximately 1 − type I fraction.
- 6. Interpret predictions in light of reference coverage, sample processing and platform differences.

This is a transcriptome-derived estimate, not a direct histological count.

Session summary and key references

Take-home messages

- Bulk RNA-seq is a mixture of signals from the cell types in a tissue.
- A single-cell atlas supplies reference signatures; a model estimates mixture weights.
- DeconRNASeq uses non-negative quadratic programming to estimate proportions.
- PredictFiberType applies this strategy to slow- and fast-twitch skeletal muscle fibres.
- Independent measurements are essential for validation.

References and resources

Gong T, Szustakowski JD. 2013. DeconRNASeq. Bioinformatics 29:1083–1085. doi:10.1093/bioinformatics/btt090

Oskolkov N et al. 2022. High-throughput muscle fiber typing from RNA sequencing data. Skeletal Muscle. doi:10.1186/s13395-022-00299-4

GitHub: OlaHanssonLab/PredictFiberType

Question: which assumptions about the reference atlas are most likely to affect an estimate?

National Bioinformatics Infrastructure Sweden (NBIS)



Knut och Alice Wallenbergs Stiftelse

Vetenskapsrådet

LUNDS UNIVERSITET

Acknowledgments: NIRI + TARGETWISE



European Regional Development Fund project “Functional Omics Analysis of Metabolic Diseases to Advance Drug Discovery Research Excellence in LIOS - TARGETWISE” (No.1.1.1.5./2/24/A/003)

2027 National Development Plan

Funded by the European Union

Acknowledgments: Metabolic Research Group



Open positions: fully covered PhD and 3-year postdoctoral fellow positions!